DOKUMENTASI IMPLEMENTASI DAN PENGUJIAN SISTEM OTOMATISASI REMEDIASI MALWARE

Integrasi Wazuh SIEM dan Shuffle SOAR pada Lingkungan IPv6

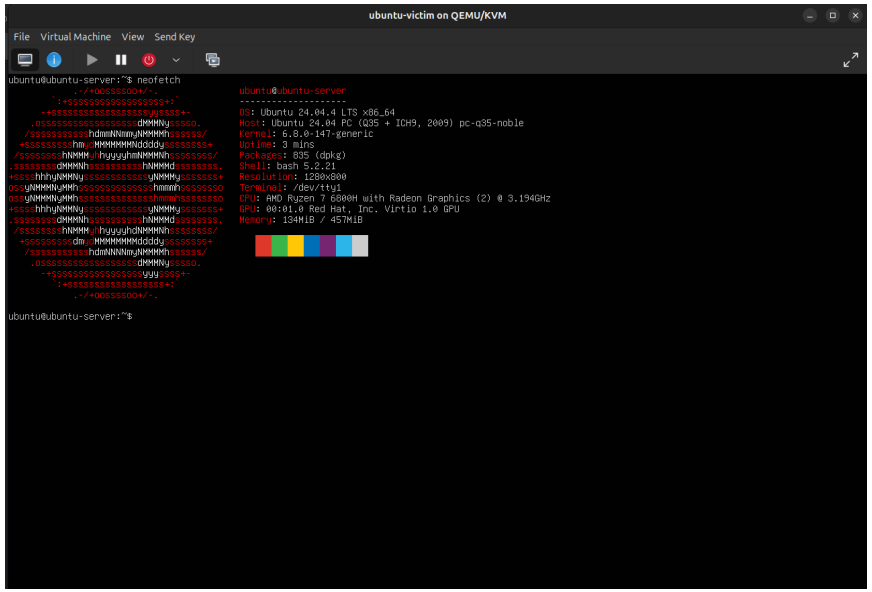
Ibrahim | NIM 3337220054 | Informatika | Universitas Sultan Ageng Tirtayasa

BAGIAN 01 · LINGKUNGAN PENELITIAN

```
NAMES                            IMAGE                                          STATUS
shuffle-frontend                 ghcr.io/shuffle/shuffle-frontend:latest        Up 3 minutes
shuffle-backend                  ghcr.io/shuffle/shuffle-backend:latest         Up 3 minutes
shuffle-orborus                  ghcr.io/shuffle/shuffle-orborus:latest         Up 3 minutes
single-node-wazuh.dashboard-1    wazuh/wazuh-dashboard:4.14.6                   Up Less than a second
single-node-wazuh.manager-1      wazuh/wazuh-manager:4.14.6                     Up Less than a second
single-node-wazuh.indexer-1      wazuh/wazuh-indexer:4.14.6                     Up Less than a second
ibrahim@ibrahim:~/Shuffle$
```

Gambar 01. Container inti Wazuh SIEM dan Shuffle SOAR yang berjalan.

Sumber: List Container Pada Docker(1).png

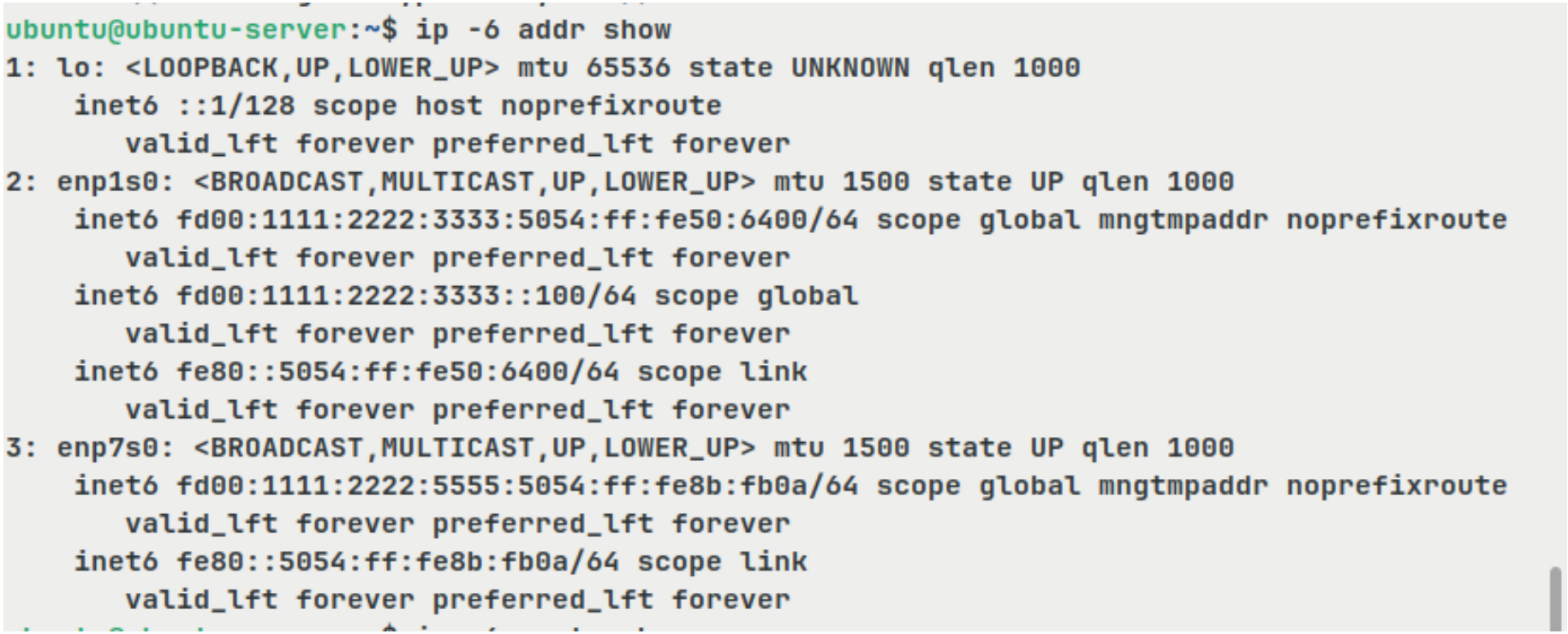


Gambar 02. Ubuntu Victim pada virtualisasi QEMU/KVM.

Sumber: Tampilan Ubuntu Victim pada QEMU KVM(1).png

BAGIAN 02 · KONFIGURASI DAN KONEKTIVITAS IPv6

Alamat IPv6 pada Ubuntu Victim



Gambar 03. Daftar alamat IPv6 pada interface Ubuntu Victim.

Sumber: Tampilan List Ipv6 Ubuntu Victim.png

BAGIAN 02 · KONFIGURASI DAN KONEKTIVITAS IPv6

Routing, ping, dan koneksi TCP/1514

```
ubuntu@ubuntu-server:~$ ip -6 route show
fd00:1111:2222:3333::/64 dev enp1s0 proto kernel metric 256 pref medium
fd00:1111:2222:3333::/64 dev enp1s0 proto ra metric 1024 mtu 1500 hoplimit 64 pref medium
fd00:1111:2222:5555::/64 dev enp7s0 proto ra metric 1024 mtu 1500 hoplimit 64 pref medium
fe80::/64 dev enp1s0 proto kernel metric 256 pref medium
fe80::/64 dev enp7s0 proto kernel metric 256 pref medium
default via fd00:1111:2222:3333::1 dev enp1s0 proto static metric 1024 pref medium
ubuntu@ubuntu-server:~$ ping -6 -c 4 fd00:1111:2222:3333::1
PING fd00:1111:2222:3333::1 (fd00:1111:2222:3333::1) 56 data bytes
64 bytes from fd00:1111:2222:3333::1: icmp_seq=1 ttl=64 time=0.333 ms
64 bytes from fd00:1111:2222:3333::1: icmp_seq=2 ttl=64 time=0.165 ms
64 bytes from fd00:1111:2222:3333::1: icmp_seq=3 ttl=64 time=0.244 ms
64 bytes from fd00:1111:2222:3333::1: icmp_seq=4 ttl=64 time=0.605 ms

--- fd00:1111:2222:3333::1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3087ms
rtt min/avg/max/mdev = 0.165/0.336/0.605/0.165 ms
ubuntu@ubuntu-server:~$ ^C
```

Gambar 04. Routing dan hasil ping IPv6 dari Ubuntu Victim.

Sumber: Tampilan Koneksi IP Route dan Ping IPv6 Ubuntu Victim.png

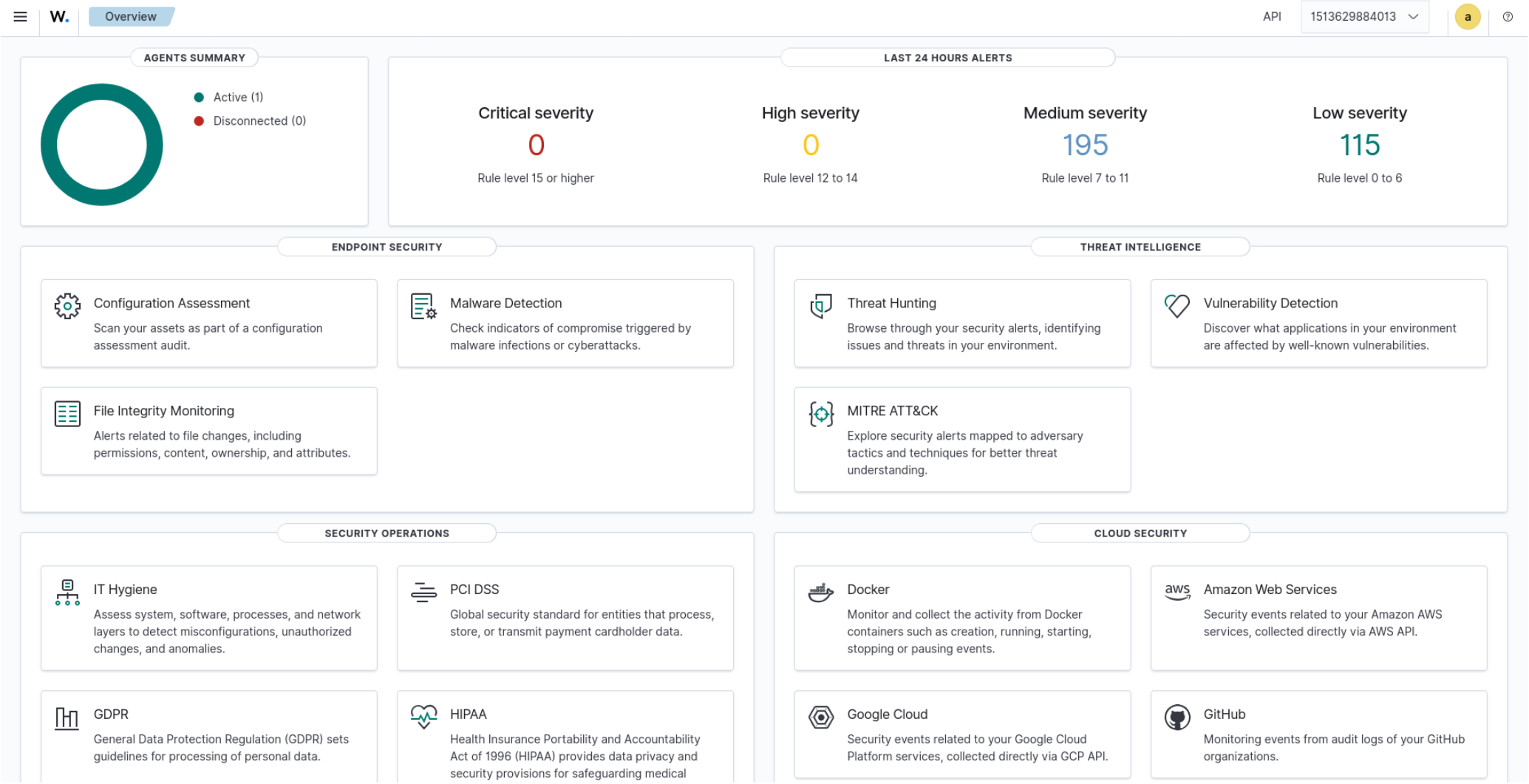
```
ubuntu@ubuntu-server:~$ nc -6 -vz -w 3 fd00:1111:2222:3333::1 1514
Connection to fd00:1111:2222:3333::1 1514 port [tcp/*] succeeded!
ubuntu@ubuntu-server:~$
```

Gambar 05. Konektivitas TCP menuju fd00:1111:2222:3333::1:1514.

Sumber: Tampilan Koneksi TCP port 1514 dengan IPv6 Ubuntu Victim.png

BAGIAN 03 · IMPLEMENTASI WAZUH SIEM

Tampilan sistem dan agent

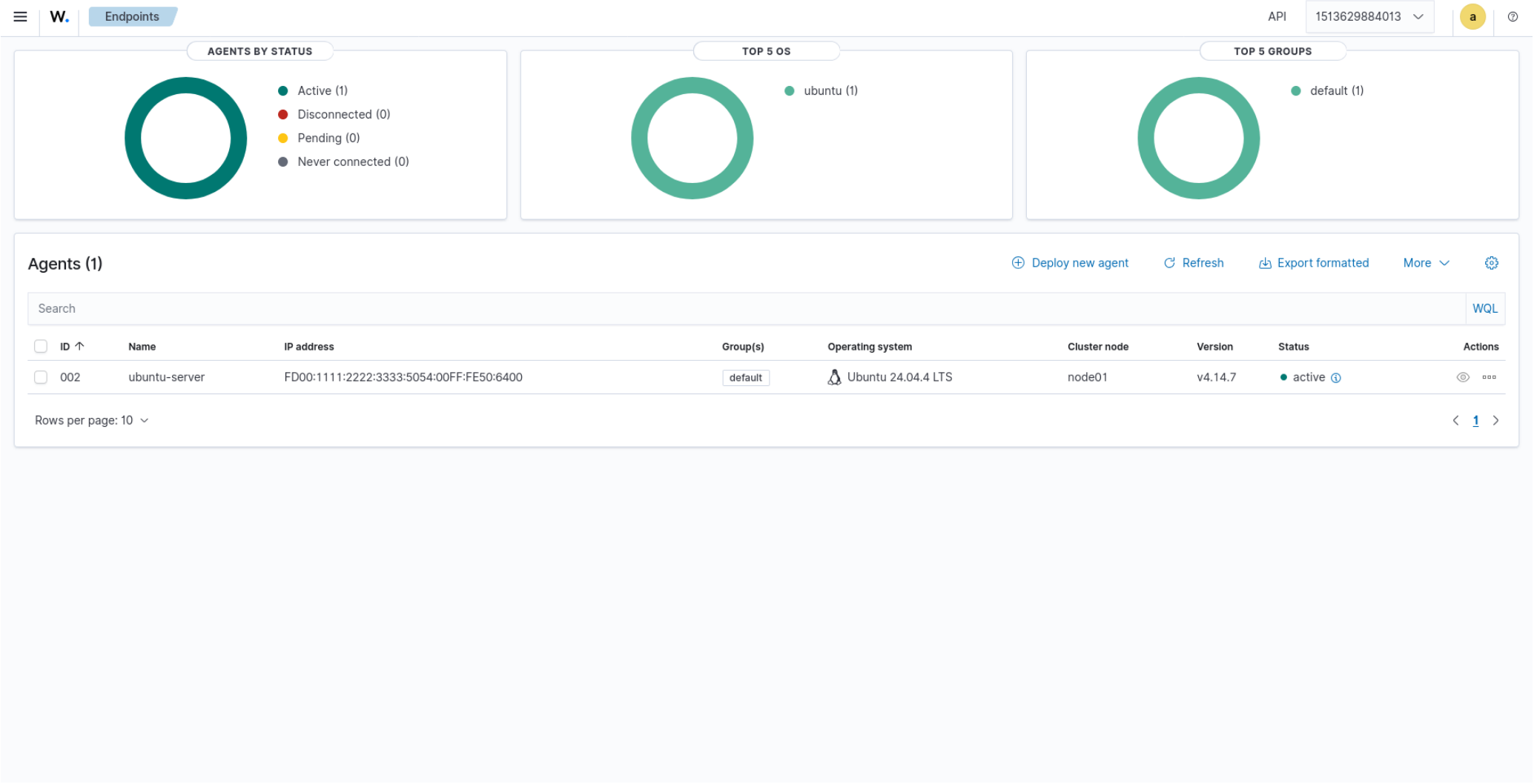


Gambar 06. Halaman overview Wazuh SIEM.

Sumber: Tampilan Overview Wazuh(1).png

BAGIAN 03 · IMPLEMENTASI WAZUH SIEM

Agent yang terhubung

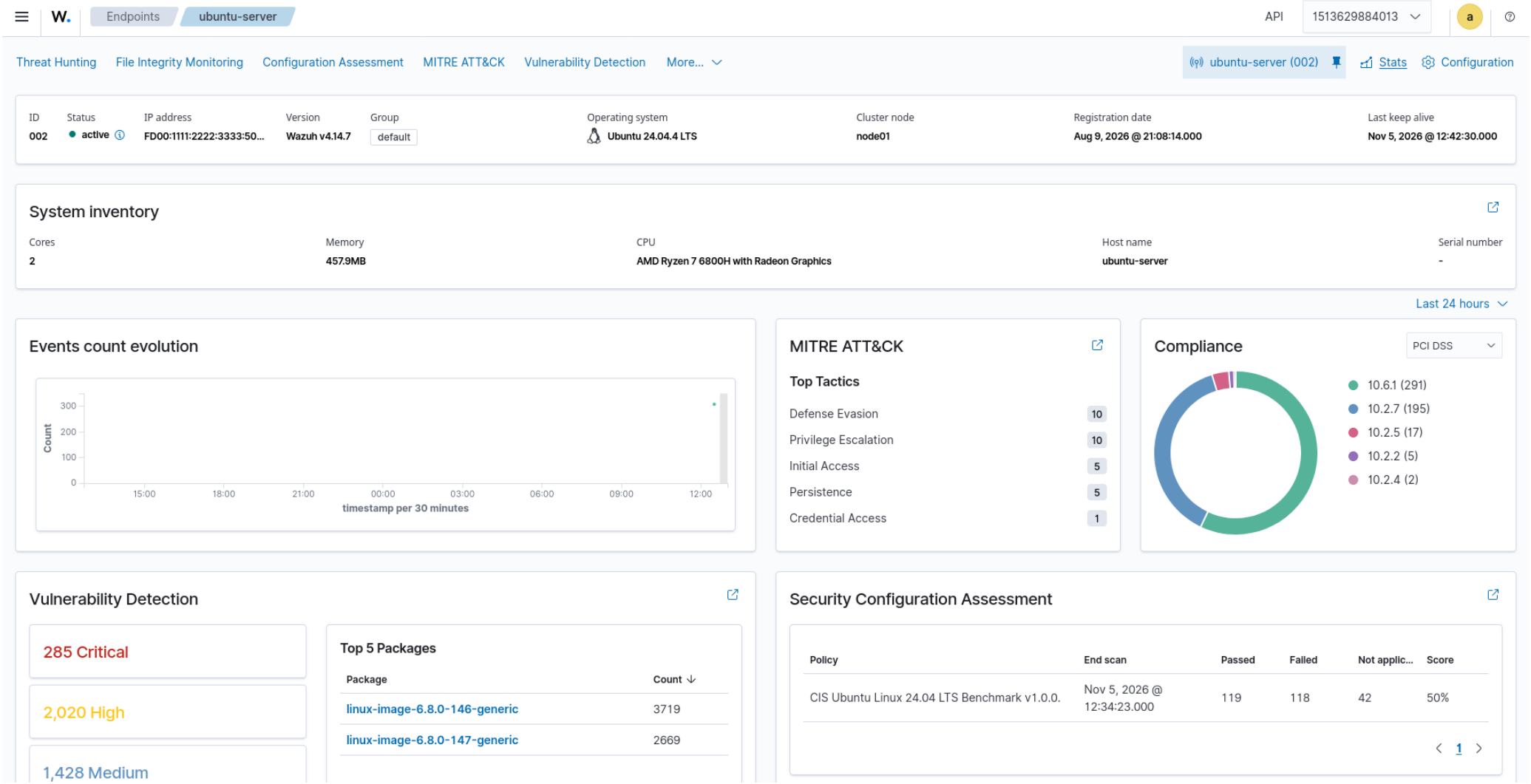


Gambar 07. Daftar Wazuh Agent dan status endpoint.

Sumber: Tampilan List Agent Wazuh (Endpoint)(1).png

BAGIAN 03 · IMPLEMENTASI WAZUH SIEM

Detail Ubuntu Victim

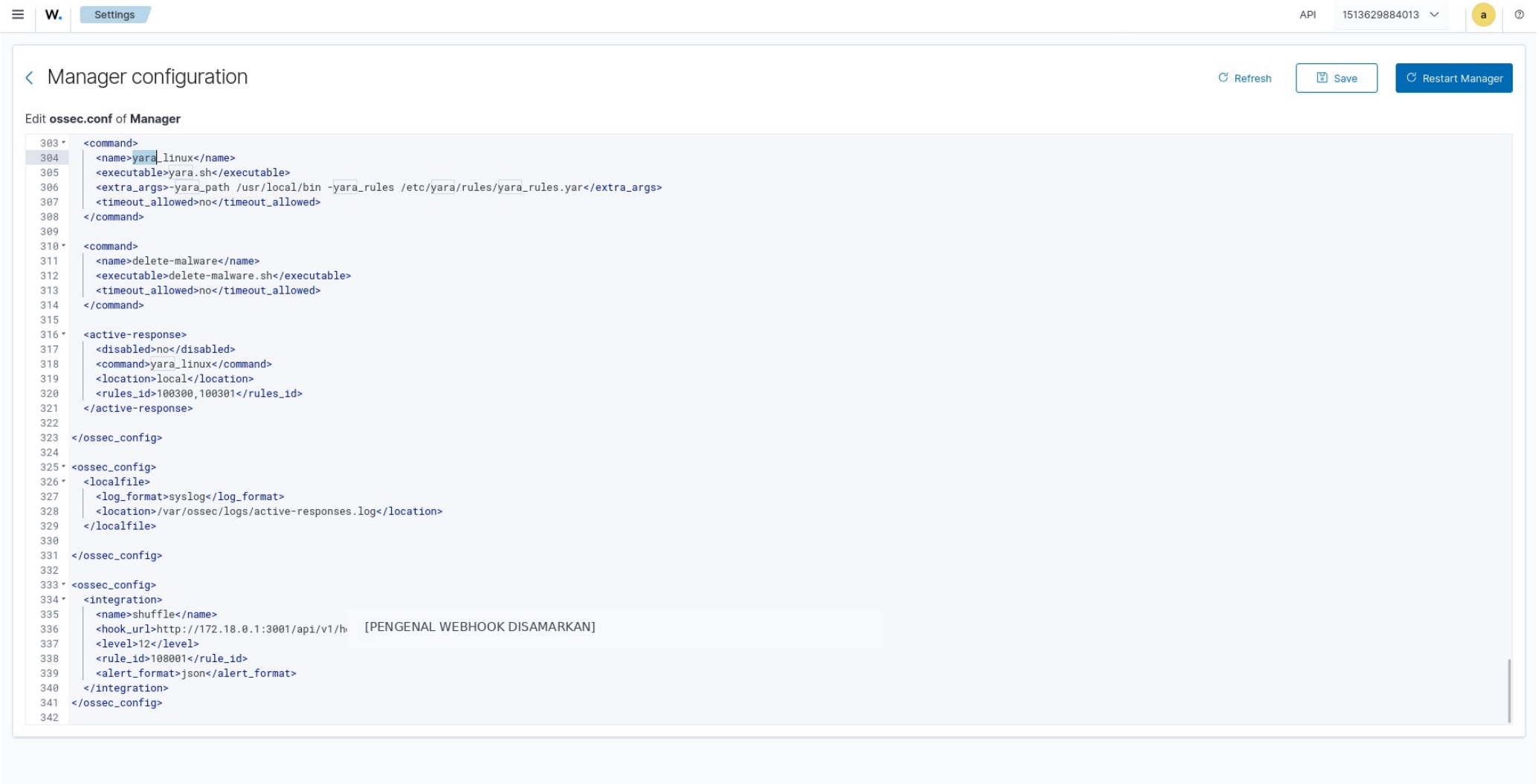


Gambar 08. Informasi endpoint ubuntu-server pada Wazuh.

Sumber: Tampilan Detail Agent Ubuntu VIctim Wazuh(1).png

BAGIAN 03 · IMPLEMENTASI WAZUH SIEM

Konfigurasi deteksi dan integrasi



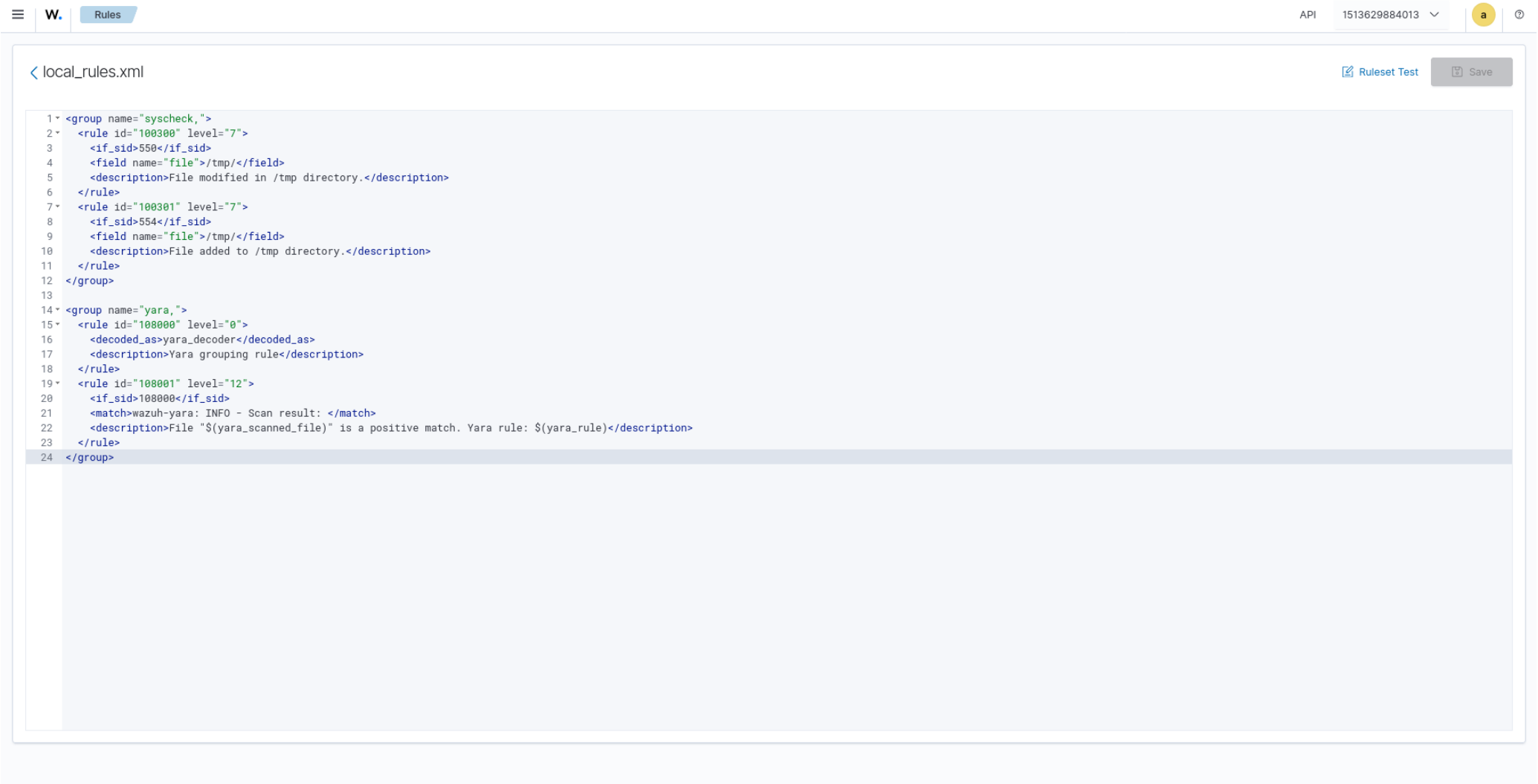
Gambar 09. Cuplikan konfigurasi ossec.conf, perintah YARA, dan Webhook.

Pengenal unik Webhook telah disamarkan.

Sumber: Tampilan Konfigurasi File ossec pada Wazuh.png

BAGIAN 03 · IMPLEMENTASI WAZUH SIEM

Custom rule Wazuh

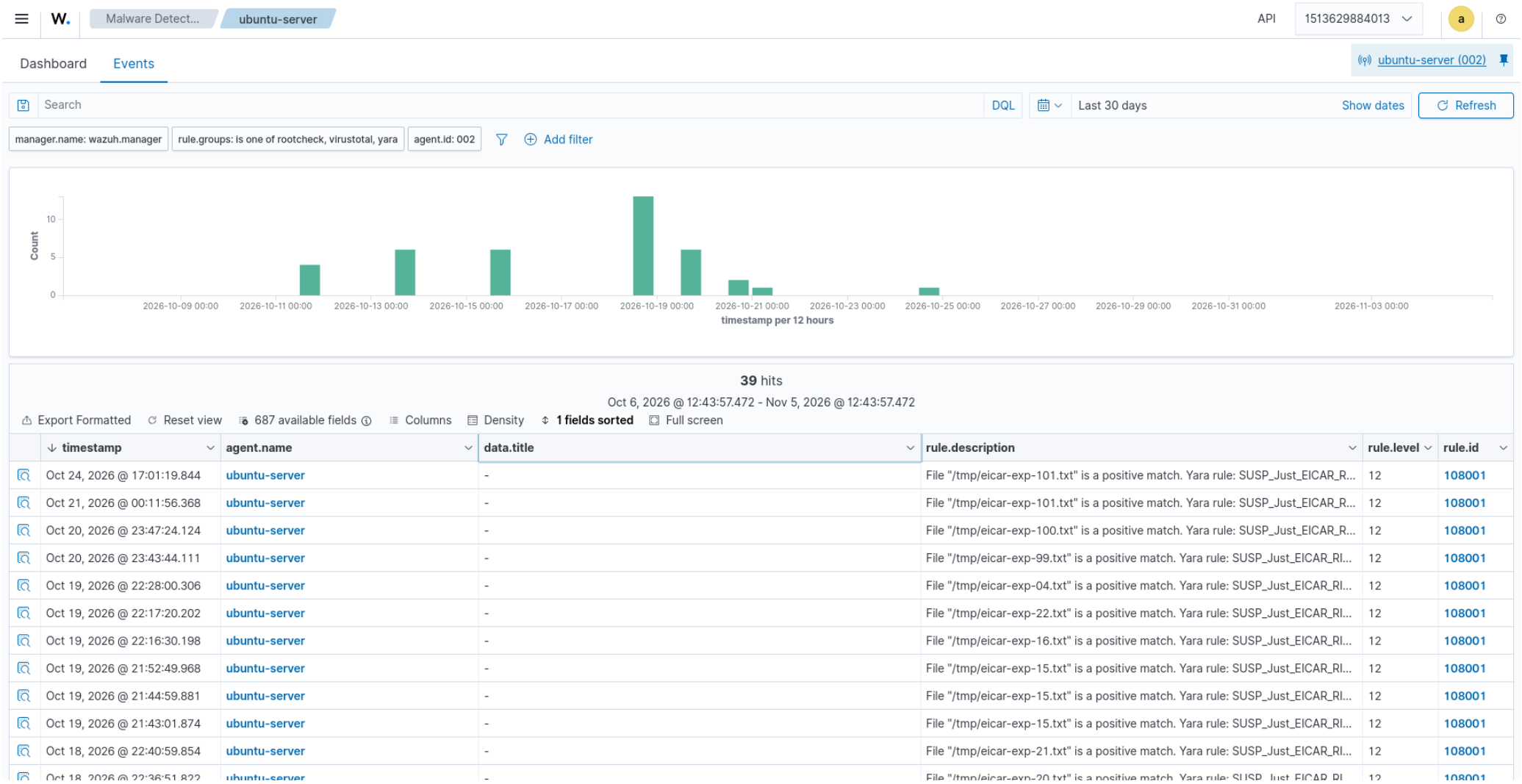


Gambar 10. Konfigurasi local_rules.xml yang digunakan.

Sumber: Tampilan local_rules xml pada Wazuh.png

BAGIAN 04 · BUKTI DETEKSI FILE EICAR

Event deteksi pada Wazuh



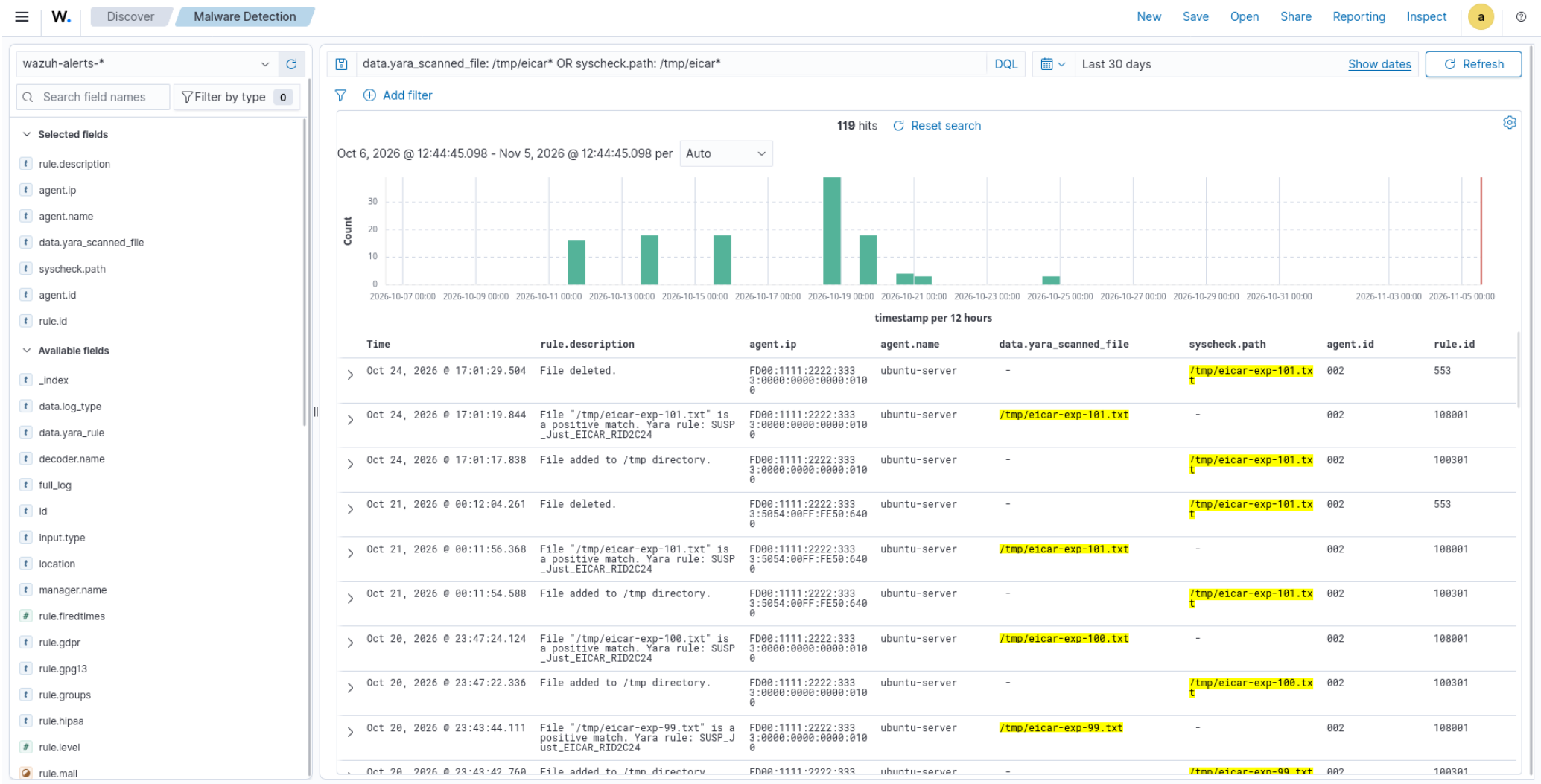
timestamp	agent.name	data.title	rule.description	rule.level	rule.id
Oct 24, 2026 @ 17:01:19.844	ubuntu-server	-	File "/tmp/eicar-exp-101.txt" is a positive match. Yara rule: SUSP_Just_EICAR_R...	12	108001
Oct 21, 2026 @ 00:11:56.368	ubuntu-server	-	File "/tmp/eicar-exp-101.txt" is a positive match. Yara rule: SUSP_Just_EICAR_R...	12	108001
Oct 20, 2026 @ 23:47:24.124	ubuntu-server	-	File "/tmp/eicar-exp-100.txt" is a positive match. Yara rule: SUSP_Just_EICAR_R...	12	108001
Oct 20, 2026 @ 23:43:44.111	ubuntu-server	-	File "/tmp/eicar-exp-99.txt" is a positive match. Yara rule: SUSP_Just_EICAR_RI...	12	108001
Oct 19, 2026 @ 22:28:00.306	ubuntu-server	-	File "/tmp/eicar-exp-04.txt" is a positive match. Yara rule: SUSP_Just_EICAR_RI...	12	108001
Oct 19, 2026 @ 22:17:20.202	ubuntu-server	-	File "/tmp/eicar-exp-22.txt" is a positive match. Yara rule: SUSP_Just_EICAR_RI...	12	108001
Oct 19, 2026 @ 22:16:30.198	ubuntu-server	-	File "/tmp/eicar-exp-16.txt" is a positive match. Yara rule: SUSP_Just_EICAR_RI...	12	108001
Oct 19, 2026 @ 21:52:49.968	ubuntu-server	-	File "/tmp/eicar-exp-15.txt" is a positive match. Yara rule: SUSP_Just_EICAR_RI...	12	108001
Oct 19, 2026 @ 21:44:59.881	ubuntu-server	-	File "/tmp/eicar-exp-15.txt" is a positive match. Yara rule: SUSP_Just_EICAR_RI...	12	108001
Oct 19, 2026 @ 21:43:01.874	ubuntu-server	-	File "/tmp/eicar-exp-15.txt" is a positive match. Yara rule: SUSP_Just_EICAR_RI...	12	108001
Oct 18, 2026 @ 22:40:59.854	ubuntu-server	-	File "/tmp/eicar-exp-21.txt" is a positive match. Yara rule: SUSP_Just_EICAR_RI...	12	108001
Oct 18, 2026 @ 22:36:51.822	ubuntu-server	-	File "/tmp/eicar-exp-20.txt" is a positive match. Yara rule: SUSP_Just_EICAR_RI...	12	108001

Gambar 11. Tampilan event deteksi pada Wazuh Malware Detection.

Sumber: Tampilan Event pada Malware Detection Wazuh(1).png

BAGIAN 04 · BUKTI DETEKSI FILE EICAR

Pencarian alert pada Discover



Gambar 12. Discover dengan filter event deteksi file EICAR.

Sumber: Tampilan Page Discover dengan filter Deteksi FIle Eicar Wazuh(1).png

BAGIAN 04 · BUKTI DETEKSI FILE EICAR

Detail FIM added (bagian 1)

Table JSON

t	_index	wazuh-alerts-4.x-2026.10.18
t	agent.id	002
t	agent.ip	FD00:1111:2222:3333:5054:00FF:FE50:6400
t	agent.name	ubuntu-server
t	decoder.name	syscheck_new_entry
t	full_log	File '/tmp/eicar-exp-17.txt' added Mode: realtime
t	id	1792336669.26875147
t	input.type	log
t	location	syscheck
t	manager.name	wazuh.manager
t	rule.description	File added to /tmp directory.
#	rule.firedtimes	2
t	rule.groups	syscheck

Gambar 13. Detail event penambahan file terpantau FIM.

Sumber: Tampilan Detail Log FIM Added File Malware - 1(1).png

BAGIAN 04 · BUKTI DETEKSI FILE EICAR

Detail FIM added (bagian 2)

◕	rule.mail	false
t	syscheck.event	added
t	syscheck.gid_after	1000
t	syscheck.gname_after	ubuntu
t	syscheck.inode_after	786453
t	syscheck.md5_after	44d88612fea8a8f36de82e1278abb02f
t	syscheck.mode	realtime
📅	syscheck.mtime_after	Oct 18, 2026 @ 22:17:49.000
t	syscheck.path	/tmp/eicar-exp-17.txt
t	syscheck.perm_after	rw-r--r--
t	syscheck.sha1_after	3395856ce81f2b7382dee72602f798b642f14140
t	syscheck.sha256_after	275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f
#	syscheck.size_after	68
t	syscheck.uid_after	1000
t	syscheck.uname_after	ubuntu

Gambar 14. Field filepath dan metadata event FIM added.

Sumber: Tampilan Detail Log FIM Added File Malware - 2(1).png

BAGIAN 04 · BUKTI DETEKSI FILE EICAR

Deteksi YARA

timestamp per 12 hours

Table JSON

t	_index	wazuh-alerts-4.x-2026.10.18
t	agent.id	002
t	agent.ip	FD00:1111:2222:3333:5054:00FF:FE50:6400
t	agent.name	ubuntu-server
t	data.log_type	INFO
t	data.yara_rule	SUSP_Just_EICAR_RID2C24
t	data.yara_scanned_file	/tmp/eicar-exp-17.txt
t	decoder.name	yara_decoder
t	full_log	wazuh-yara: INFO - Scan result: SUSP_Just_EICAR_RID2C24 /tmp/eicar-exp-17.txt
t	id	1792336671.26875665
t	input.type	log
t	location	/var/ossec/logs/active-responses.log
t	manager.name	wazuh.manager
t	rule.description	File "/tmp/eicar-exp-17.txt" is a positive match. Yara rule: SUSP_Just_EICAR_RID2C24
#	rule.firedtimes	1
t	rule.groups	yara
t	rule.id	108001
#	rule.level	12

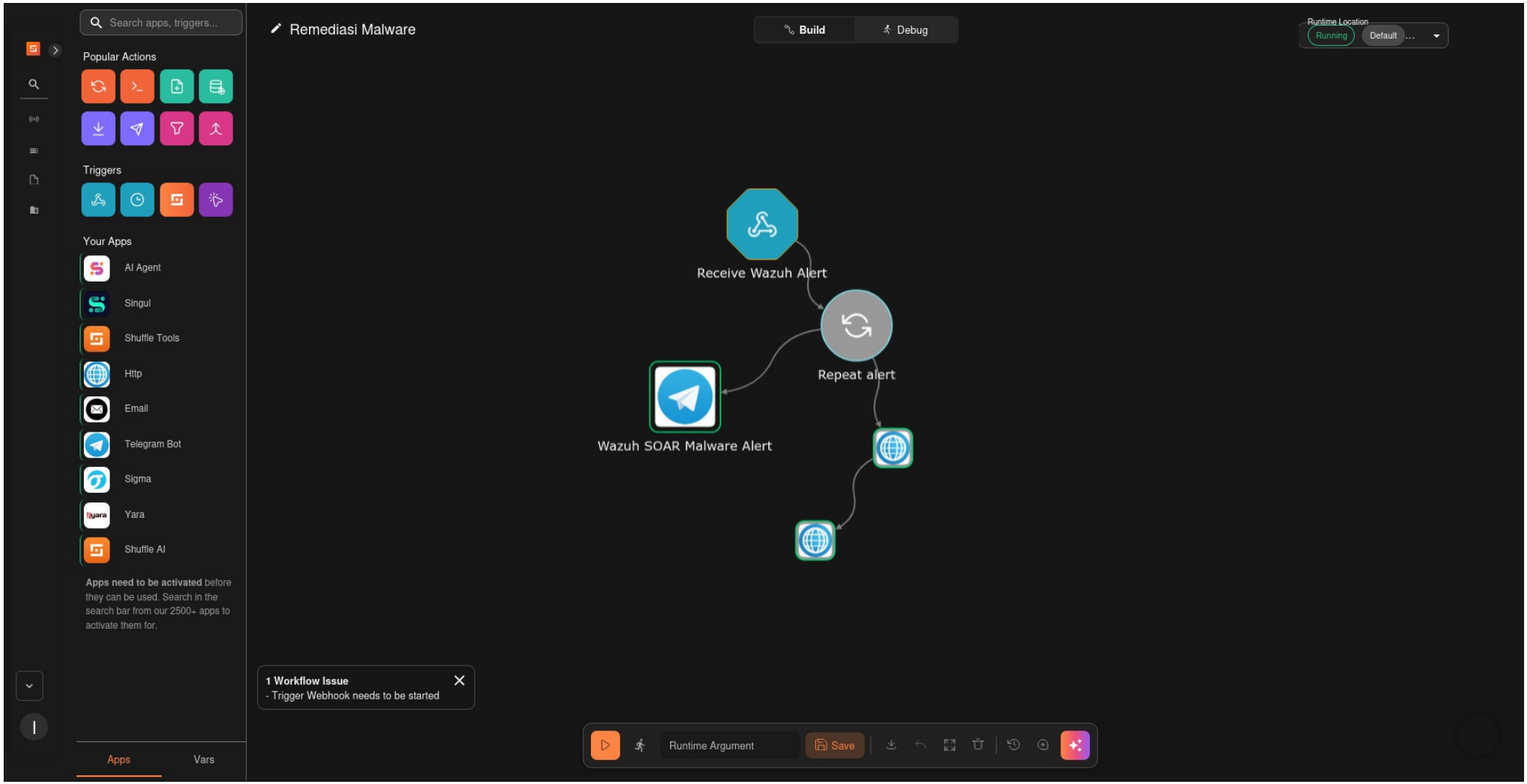
Gambar 15. Alert YARA terhadap /tmp/eicar-exp-17.txt, Agent 002, Rule 108001.

Dokumentasi event Oktober 2026; tidak disamakan dengan eksekusi Shuffle tanggal lain.

Sumber: Tampilan Detail Alert Deteksi Malware oleh YARA(1).png

BAGIAN 05 · IMPLEMENTASI SHUFFLE SOAR

Rancangan workflow remediasi

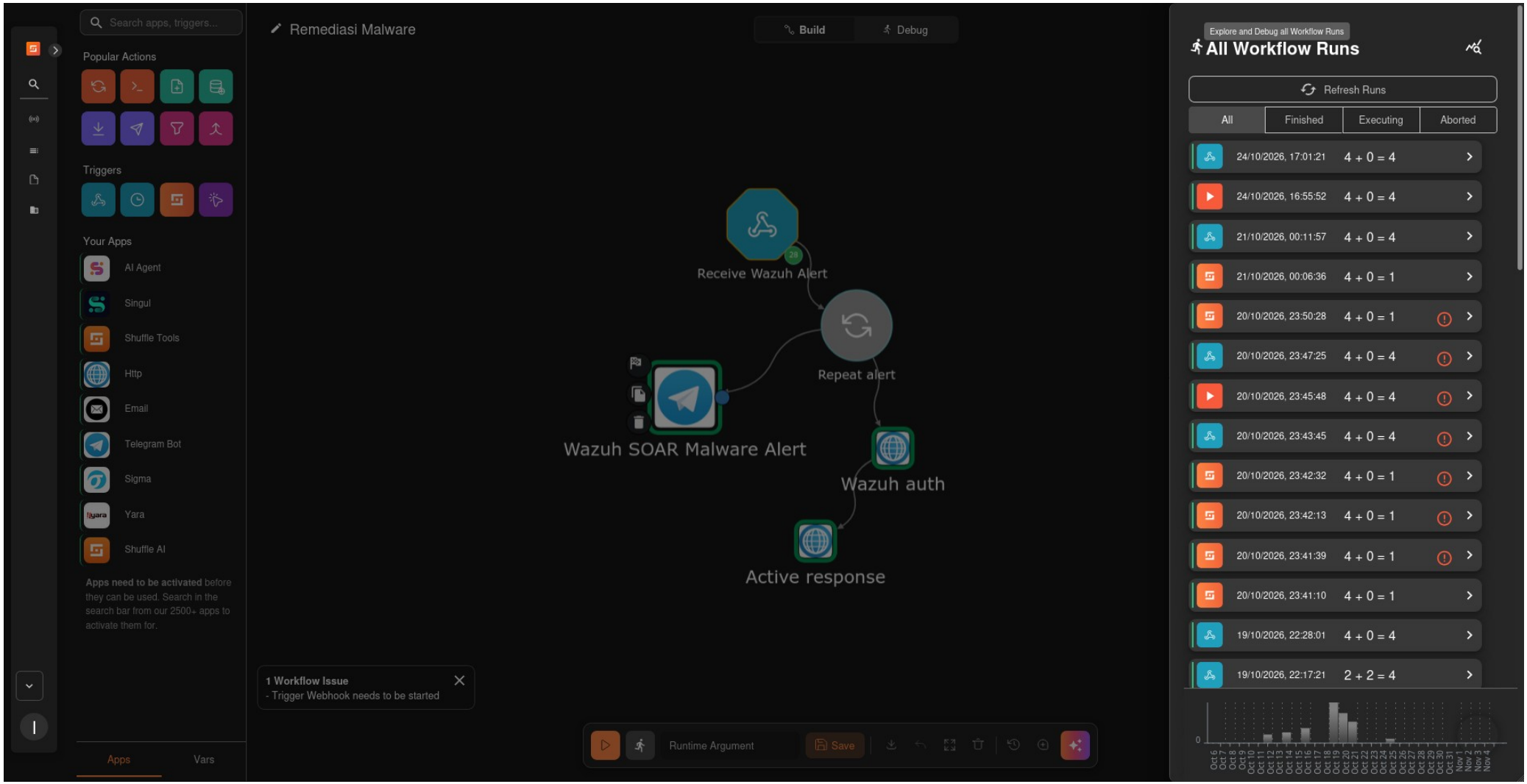


Gambar 16. Workflow remediasi malware pada Shuffle SOAR.

Sumber: Tampilan Workflow Remediasi Malware Shuffle SOAR(1).png

BAGIAN 05 · IMPLEMENTASI SHUFFLE SOAR

Riwayat eksekusi workflow

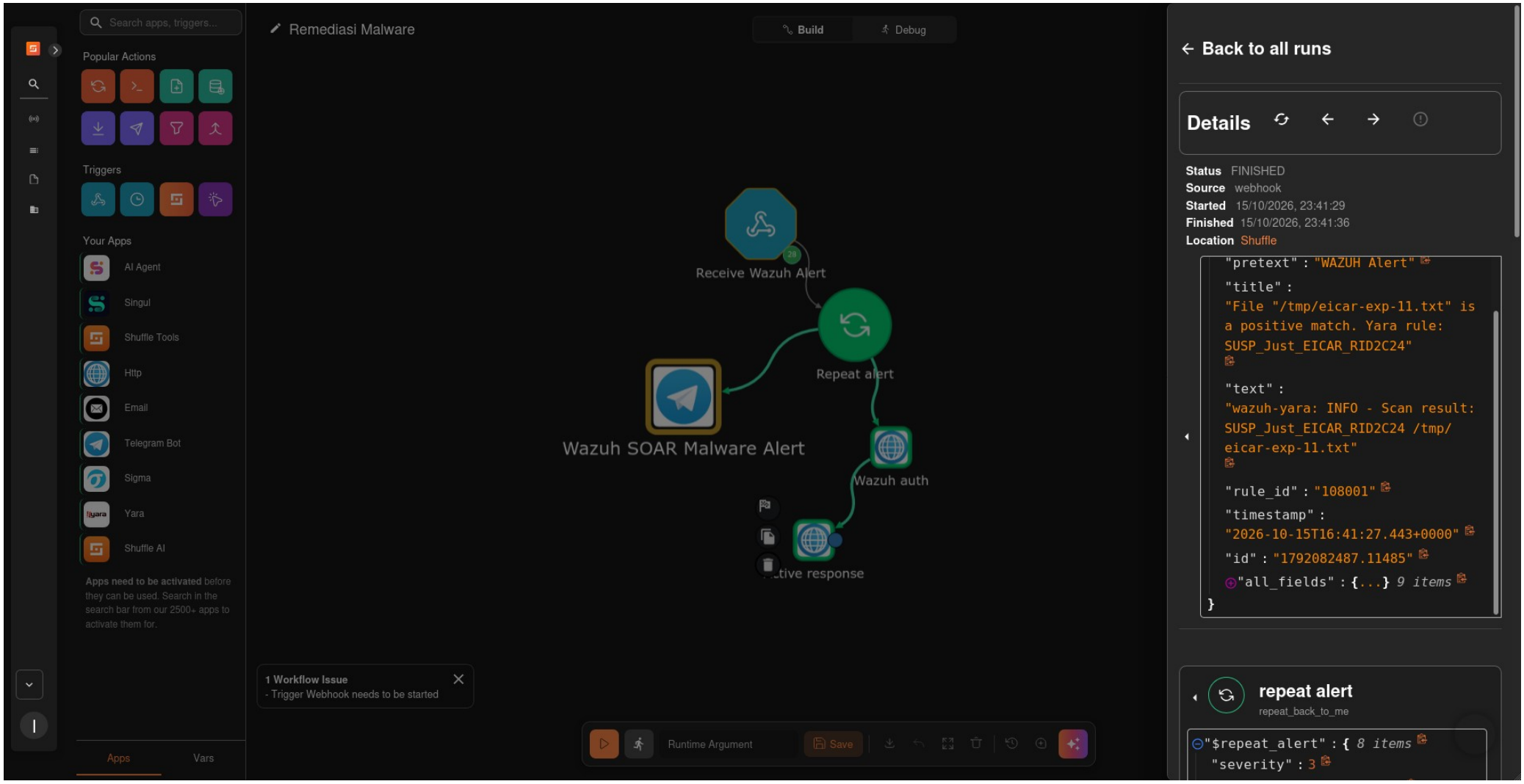


Gambar 17. Riwayat eksekusi workflow Shuffle.

Sumber: Tampilan ALL Workflow Runs Shuffle SOAR(1).png

BAGIAN 05 · IMPLEMENTASI SHUFFLE SOAR

Detail eksekusi workflow

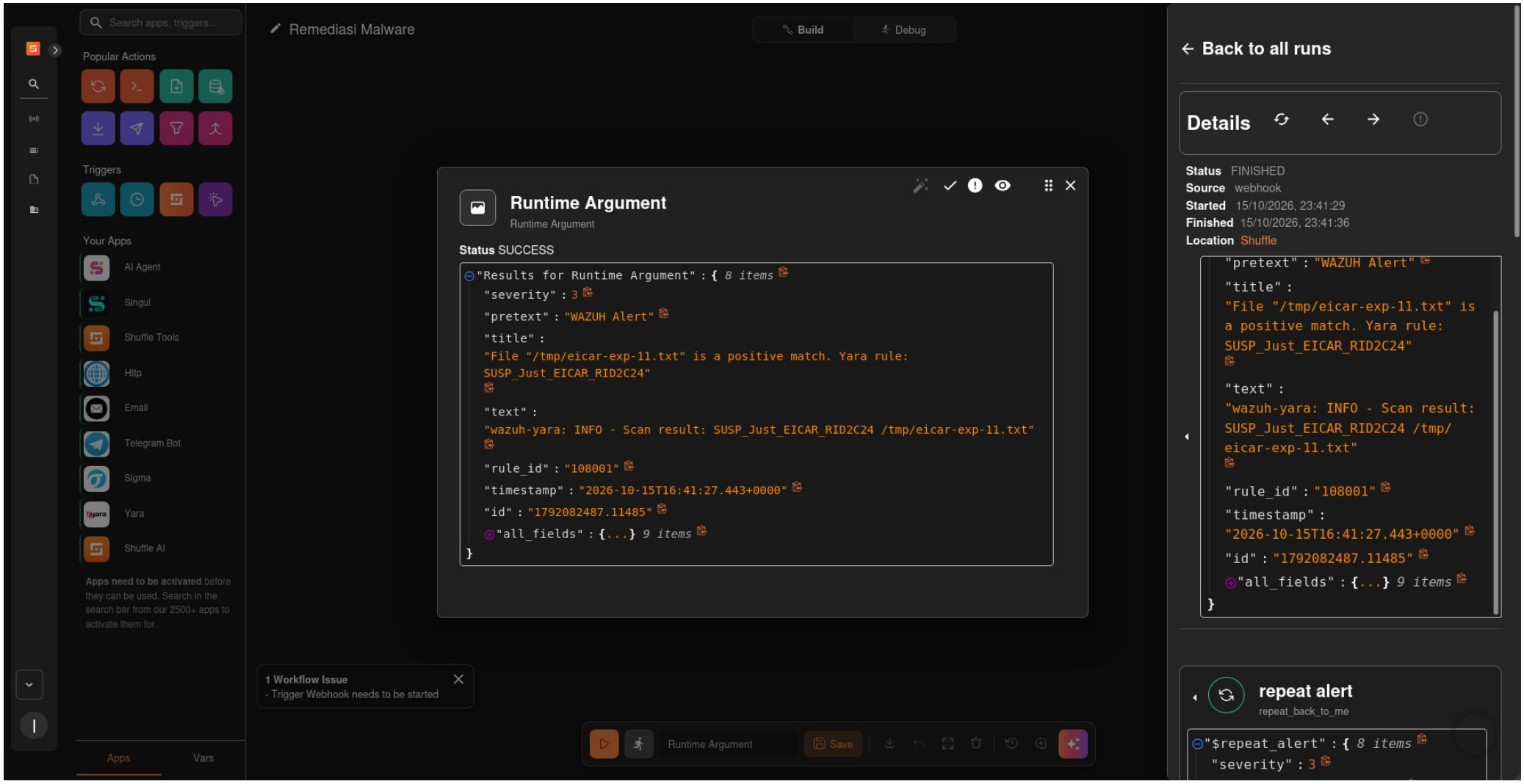


Gambar 18. Status dan node pada salah satu eksekusi workflow.

Sumber: Tampilan isi salah satu workflow yang di run pada shuffle soar(1).png

BAGIAN 05 · IMPLEMENTASI SHUFFLE SOAR

Runtime argument



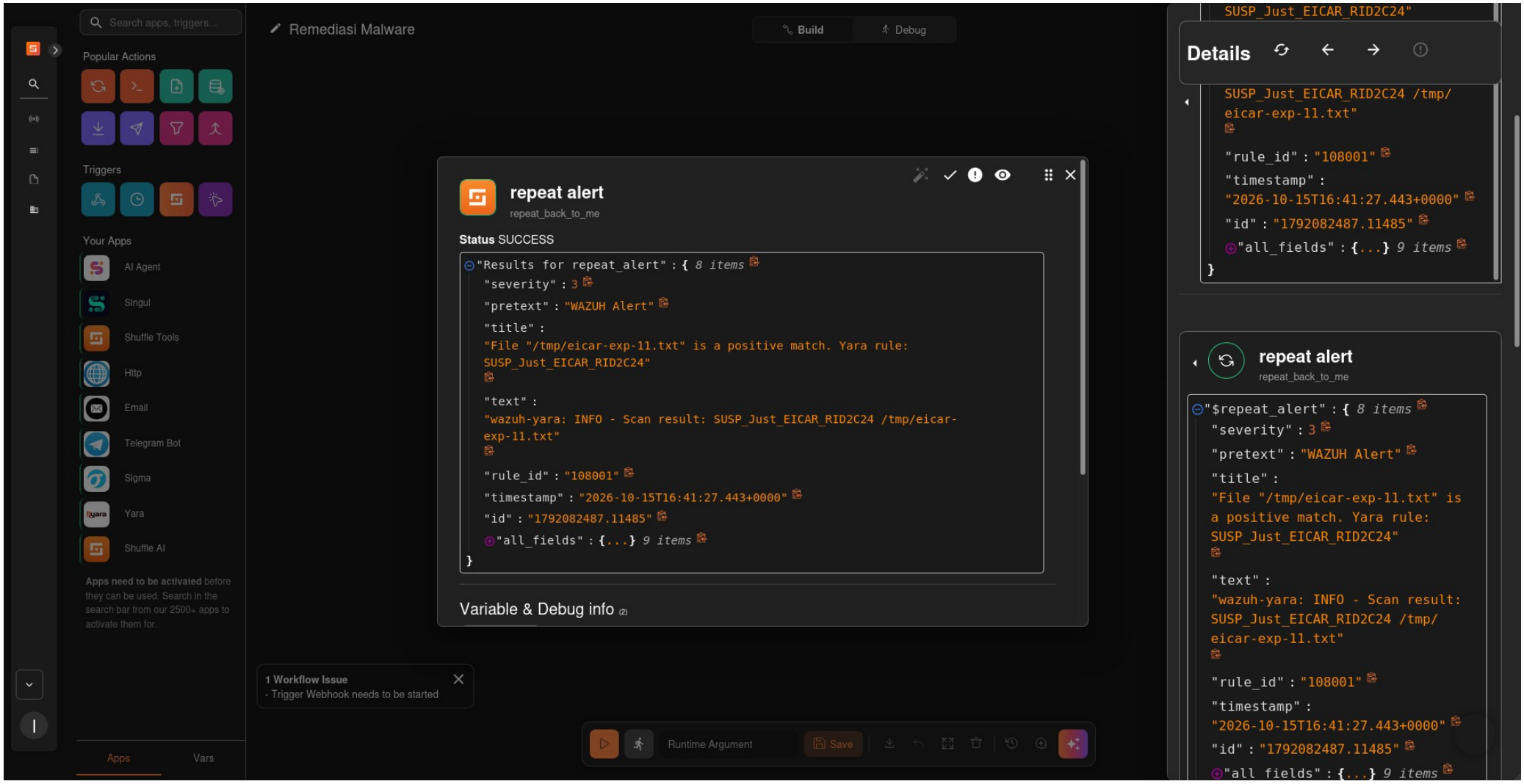
Gambar 19. Runtime argument dengan identitas /tmp/eicar-exp-11.txt.

Bukti episode lain; tidak digabung sebagai satu kejadian dengan /tmp/eicar-exp-17.txt.

Sumber: Tampilan Runtime Argument Workflow Shuffle SOAR(1).png

BAGIAN 05 · IMPLEMENTASI SHUFFLE SOAR

Node pemrosesan alert

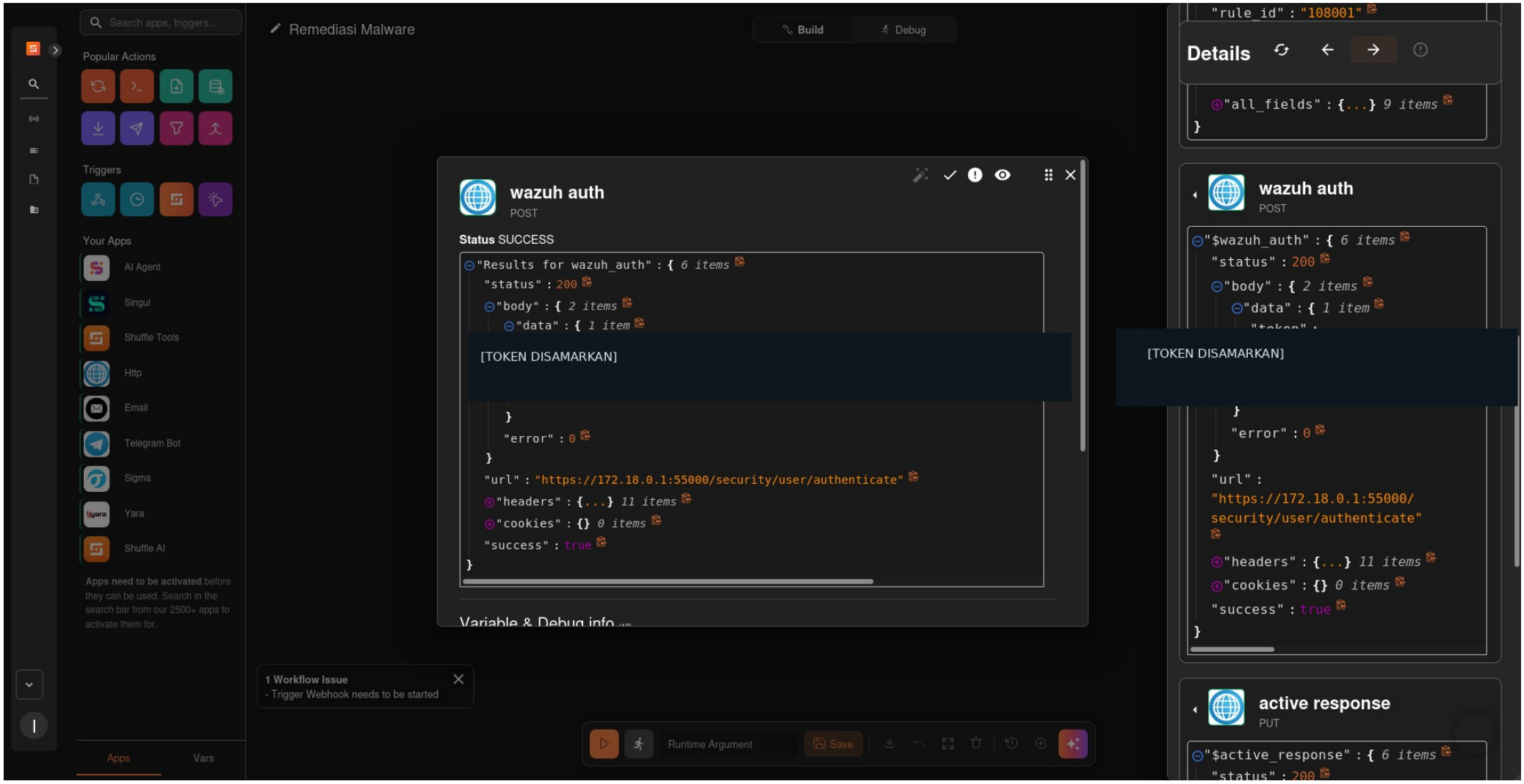


Gambar 20. Node repeat alert dan data yang diteruskan.

Sumber: Tampilan Node Repeat Alert Workflow Shuffle SOAR.png

BAGIAN 05 · IMPLEMENTASI SHUFFLE SOAR

Autentikasi Wazuh API



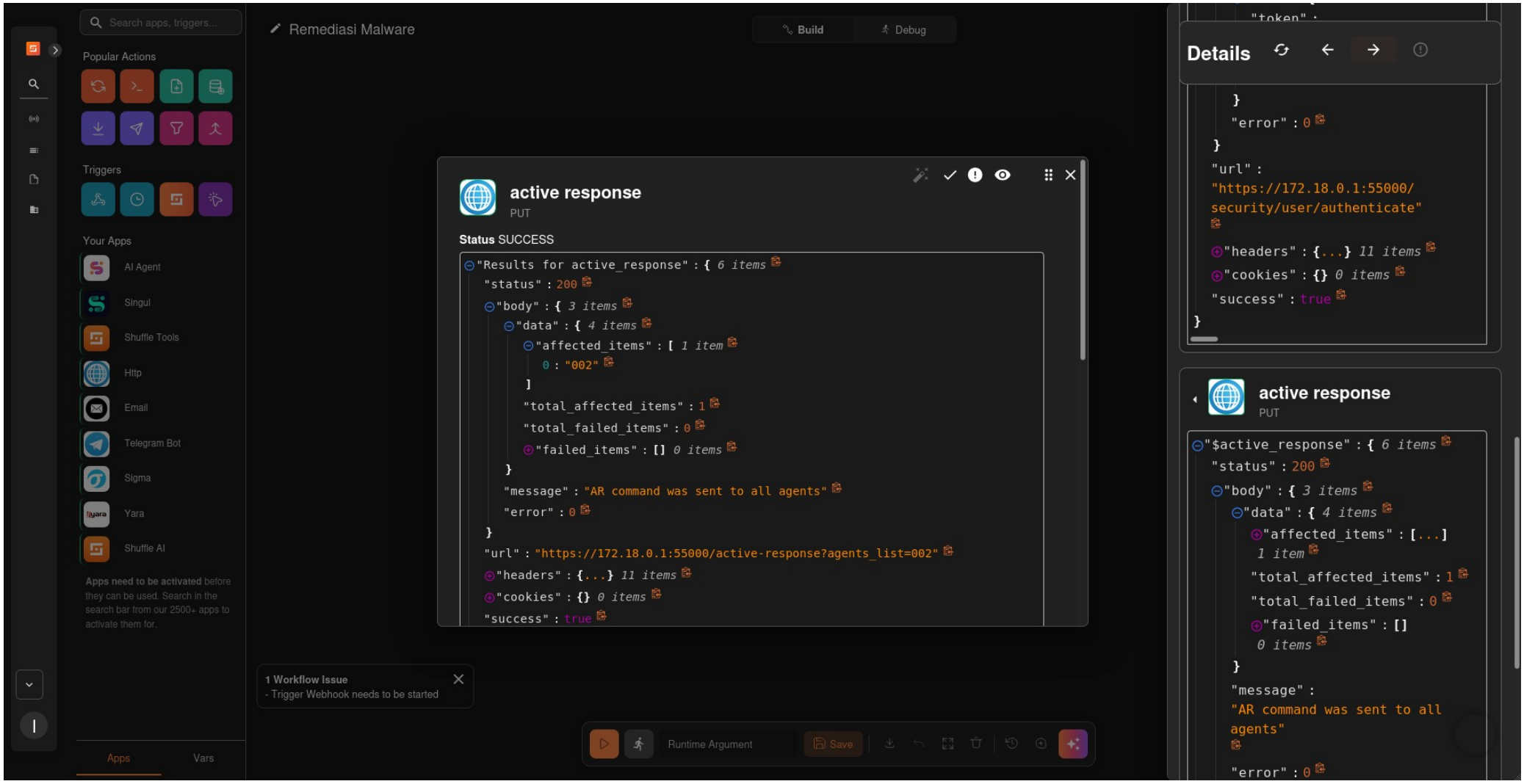
Gambar 21. Status autentikasi Wazuh API (HTTP 200).

Nilai token autentikasi disamarkan.

Sumber: Tampilan Node Wazuh Authenticate Workflow Shuffle SOAR.png

BAGIAN 05 · IMPLEMENTASI SHUFFLE SOAR

Pengiriman Active Response



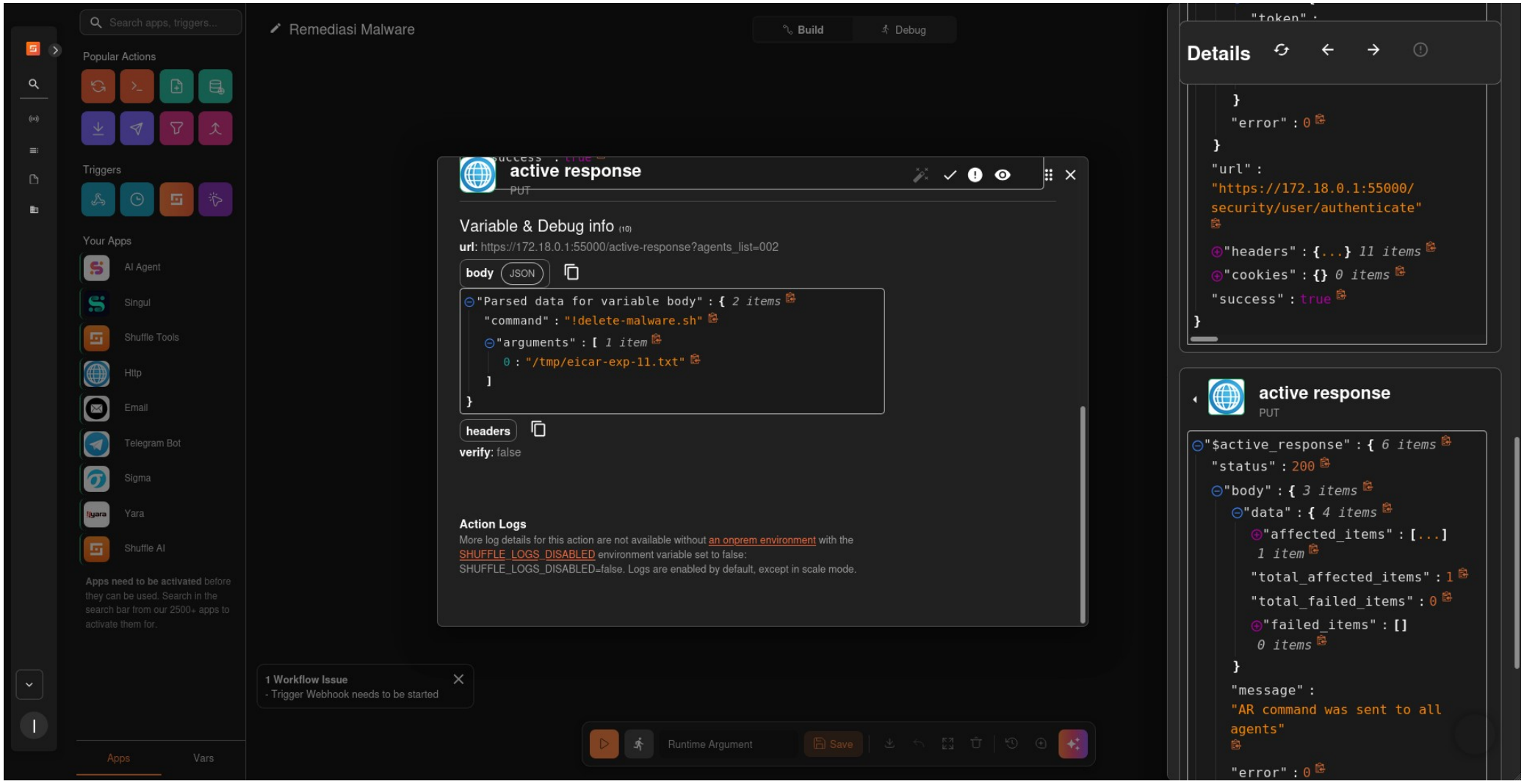
Gambar 22. Respons API pengiriman Active Response ke Agent 002.

HTTP 200 mengonfirmasi pengiriman perintah, bukan kondisi akhir file.

Sumber: Tampilan Node Wazuh Active Response Workflow Shuffle SOAR.png

BAGIAN 05 · IMPLEMENTASI SHUFFLE SOAR

Request ke Wazuh API



Gambar 23. Request body !delete-malware.sh dan filepath sasaran.

Sumber: Tampilan Body Pada Request yang dikirim oleh Node Active Response Workflow Shuffle SOAR.png

BAGIAN 06 · BUKTI PENGHAPUSAN FILE

Alert FIM deleted (bagian 1)

t	_index	wazuh-alerts-4.x-2026.10.18
t	agent.id	002
t	agent.ip	FD00:1111:2222:3333:5054:00FF:FE50:6400
t	agent.name	ubuntu-server
t	decoder.name	syscheck_deleted
t	full_log	File '/tmp/eicar-exp-17.txt' deleted Mode: realtime
t	id	1792336741.26877054
t	input.type	log
t	location	syscheck
t	manager.name	wazuh.manager
t	rule.description	File deleted.
#	rule.firedtimes	2
t	rule.gdpr	II_5.1.f
t	rule.gpg13	4.11
t	rule.groups	ossec, syscheck, syscheck_entry_deleted, syscheck_file
t	rule.hipaa	164.312.c.1, 164.312.c.2
t	rule.id	553

Gambar 24. Event FIM deleted pada file EICAR.

Sumber: Tampilan Detail Log FIM Deleted File EICAR - 1(1).png

BAGIAN 06 · BUKTI PENGHAPUSAN FILE

Alert FIM deleted (bagian 2)

t	rule.mitre.tactic	Defense Evasion, Impact
t	rule.mitre.technique	File Deletion, Data Destruction
t	rule.nist_800_53	SI.7
t	rule.pci_dss	11.5
t	rule.tsc	PI1.4, PI1.5, CC6.1, CC6.8, CC7.2, CC7.3
t	syscheck.event	deleted
t	syscheck.gid_after	1000
t	syscheck.gname_after	ubuntu
t	syscheck.inode_after	786453
t	syscheck.md5_after	44d88612fea8a8f36de82e1278abb02f
t	syscheck.mode	realtime
🗓	syscheck.mtime_after	Oct 18, 2026 @ 22:17:49.000
t	syscheck.path	/tmp/eicar-exp-17.txt
t	syscheck.perm_after	rw-r--r--
t	syscheck.sha1_after	3395856ce81f2b7382dee72602f798b642f14140
t	syscheck.sha256_after	275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f
#	syscheck.size_after	68
t	syscheck.uid_after	1000

Gambar 25. Metadata tambahan event FIM deleted.

Sumber: Tampilan Detail Log FIM Deleted File EICAR - 2(1).png

BAGIAN 06 · BUKTI EKSEKUSI ACTIVE RESPONSE

Log asli: active-responses-tail.log

Cuplikan log historis • 11 Oktober 2026 • /tmp/eicar.com

```
2026/10/11 14:03:48 delete-malware: INFO - Target diterima: /tmp/eicar.com
2026/10/11 14:03:48 delete-malware: SUCCESS - File deleted: /tmp/eicar.com
2026/10/11 15:13:06 delete-malware: INFO - Target diterima: /tmp/eicar.com
2026/10/11 15:13:06 delete-malware: SUCCESS - File deleted: /tmp/eicar.com
2026/10/11 15:17:16 delete-malware: INFO - Target diterima: /tmp/eicar.com
2026/10/11 15:17:16 delete-malware: SUCCESS - File deleted: /tmp/eicar.com
```

Sumber: Library /SKRIPSI/active-responses-tail.log (baris asli yang dipilih).
Cuplikan ini bukan log FUNC-E2E-01 tanggal 24 Oktober dan tidak dikorelasikan langsung dengan screenshot file /tmp/eicar-exp-17.txt.

BAGIAN 06 · VERIFIKASI FUNGSIONAL TAMBAHAN

FUNC-E2E-01 · 24 Oktober 2026 · File /tmp/eicar-exp-101.txt

Cuplikan asli log Active Response dari arsip FUNC-E2E-01

```
wazuh-yara: INFO - Scan result: SUSP_Just_EICAR_RID2C24 /tmp/eicar-exp-101.txt
2026/10/24 10:01:31 delete-malware: INFO - Target diterima: /tmp/eicar-exp-101.txt
2026/10/24 10:01:31 delete-malware: SUCCESS - File deleted: /tmp/eicar-exp-101.txt
```

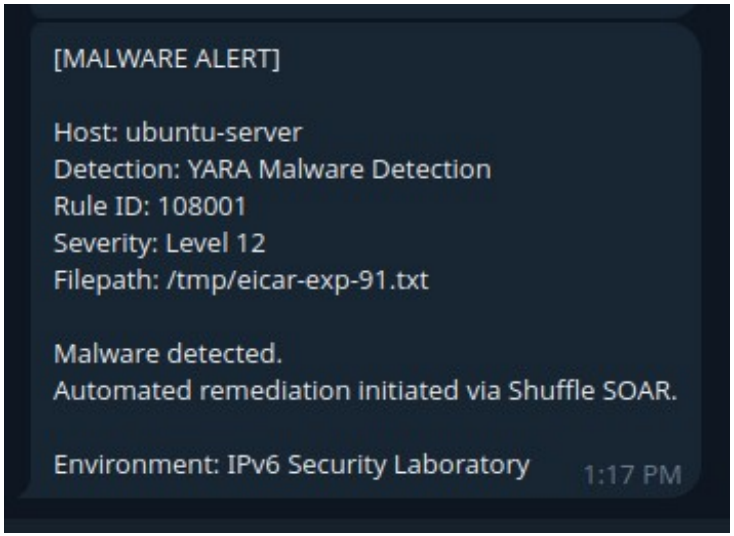
Pemeriksaan filesystem dari arsip FUNC-E2E-01

```
ubuntu@ubuntu-server:~$ test ! -e /tmp/eicar-exp-101.txt && echo "FILE_ABSENT" || echo "FILE_STILL_PRESENT"
FILE_ABSENT
```

Sumber: Pengujian End to End Workflow Remediasi(3).txt, bukti tambahan di luar file Library.
Catatan: pada FUNC-E2E-01, timestamp event FIM deleted tercatat lebih awal daripada log keberhasilan script; urutan persis antar-sumber belum dapat dipastikan.

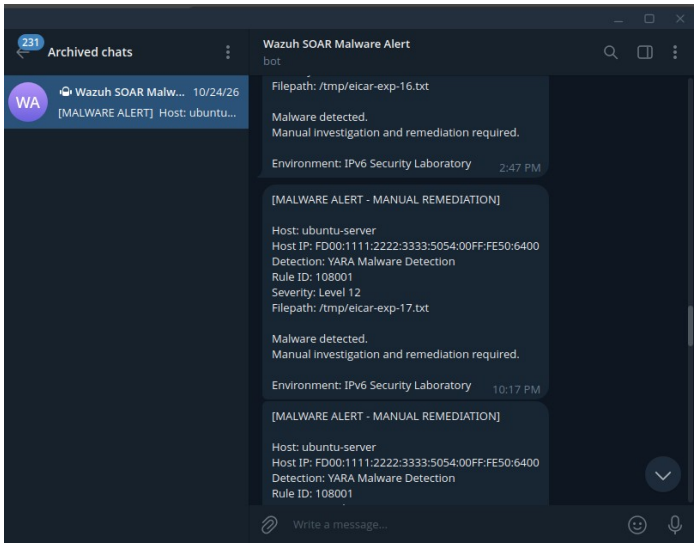
BAGIAN 07 · NOTIFIKASI TELEGRAM

Notifikasi sebagai fitur pendukung



Gambar 26. Notifikasi Telegram yang menampilkan Rule ID, severity, dan filepath.

Sumber: Tampilan Pesan Notifikasi Telegram.png



Gambar 27. Riwayat notifikasi dari integrasi Shuffle ke Telegram.

Sumber: Tampilan Notifikasi Telegram.png

Telegram merupakan fitur pendukung, bukan bukti keberhasilan penghapusan dan tidak digunakan pada seluruh pengujian utama.